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ARTICLES

Paleomagnetic and $^{40}Ar/^{39}Ar$ Study of the Sinyai Dolerite, Kenya: Implications for Gondwana Assembly[1]

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ABSTRACT

New paleomagnetic and $^{40}Ar/^{39}Ar$ age data from the Sinyai metadolerite dike in central Kenya support the suggestion that the eastern portion of Gondwana was assembled during two separate orogenic events. The dike intrudes Mozambique Belt metasediments dated ≈700 Ma and was itself metamorphosed to greenschist facies at 547 ± 4 Ma. This greenschist-facies event reset the original magnetization in the rocks and occurred over a time span that included at least one field reversal. The paleomagnetic pole at 20°S, 319°E ($\delta p = 3°$, $\delta m = 5°$) augments the available paleomagnetic database for Gondwana and suggests that Gondwana assembly was completed by 550 Ma; therefore the concept of a united East Gondwana continent may not be valid for pre-550 Ma time. In our model, the 650–800 Ma East Africa Orogeny resulted from a collision between the Congo craton of East Africa and the IMSLEK terranes (India, Madagascar, Sri Lanka, Enderby Land, and the Kalahari craton). A pervasive granulate-facies metamorphis event at ≈550 Ma in parts of East Gondwana, coupled with our paleomagnetic evidence for a united Gondwana at 550 Ma, led to our suggestion of a Kuunga Orogeny at this time. The Kuunga Orogeny results from the collision of Australo-Antarctica with Congo-IMSLEK.

Introduction

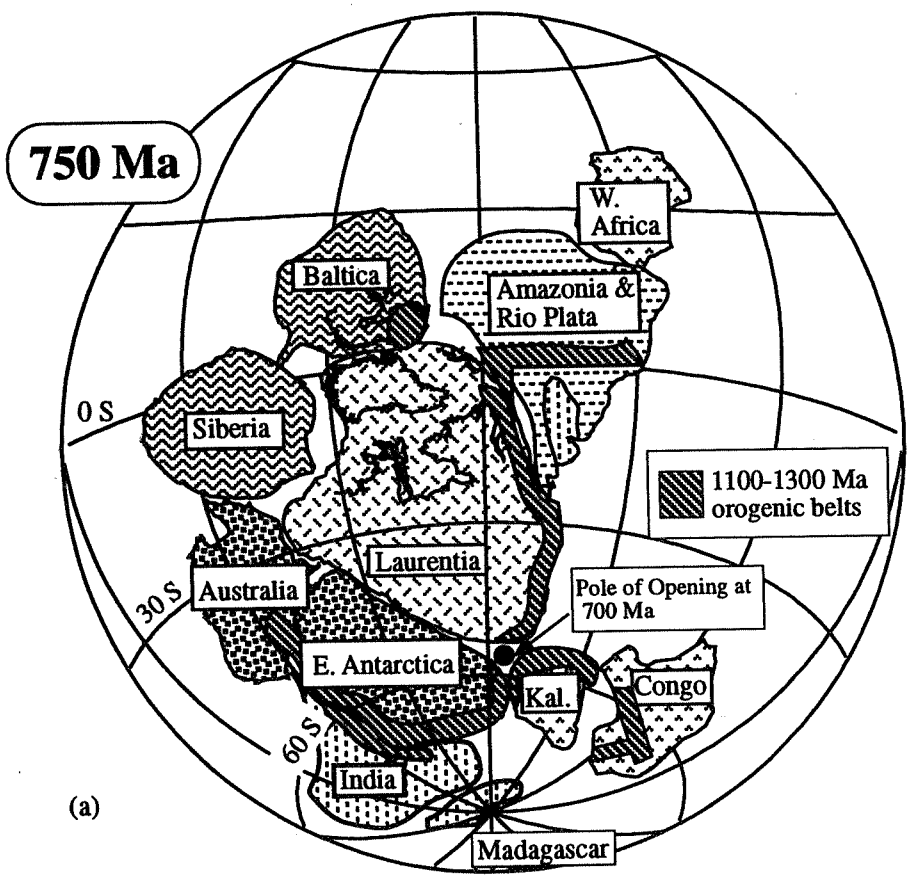
The timing and sequence of orogenic events leading to the formation of Gondwana is much debated. The debate stems, in part, from a lack of well-dated paleomagnetic poles from the various cratons that comprise greater Gondwana and uncertainties about their precollisional configurations. Recent compilations of geochronologic data from metamorphic belts within Gondwana indicate a multiplicity of discrete orogenic events between the nuclei rather than a prolonged, "Pan-African orogenic cycle" (Stern 1994; Shiriashi et al. 1994; Harris and Beeson 1993; Kröner 1993). The available geochronologic and paleomagnetic data have led to two significantly different broad categories of models for the timing of Gondwana assembly. In the first group, the 650–800 Ma East Africa Orogen that formed the Mozambique Belt (figure 1) represents the suturing of East Gondwana (India, Antarctica and Australia) and West Gondwana (Africa and South America) and the 550 Ma ages found throughout Gondwana reflect minor adjustments within the already assembled supercontinent (Stern 1994; Unrug 1992; Burke and Dewey 1972). The second group views the assembly of Gondwana as a Cambro-Ordovician event with relatively minor significance attached to the chronologic data supporting an older East Africa Orogeny (Li and Powell 1993; Powell et al. 1993; Piper 1987; McWilliams 1981). Indeed, a second debate arises within these two groups in regard to the pieces colliding to form greater Gondwana. Were East and West Gondwana internally coherent prior to their collision, or did Gondwana assembly result from the amalgamation of formerly distinct nuclei? These questions will ultimately be answered by detailed paleomagnetic, geochronologic, and structural data from the various Gondwana blocks.

Of particular importance to this debate is detailed knowledge about the breakup of the preceding supercontinent Rodinia (figure 1*a*; Dalziel 1992) because its final dispersal appears to be more or less synchronous with the assembly of Gondwana. The Rodinia configuration is considered valid from 1100 to 750 Ma. According to Dalziel (1992), by 750 Ma evidence for the first rifting

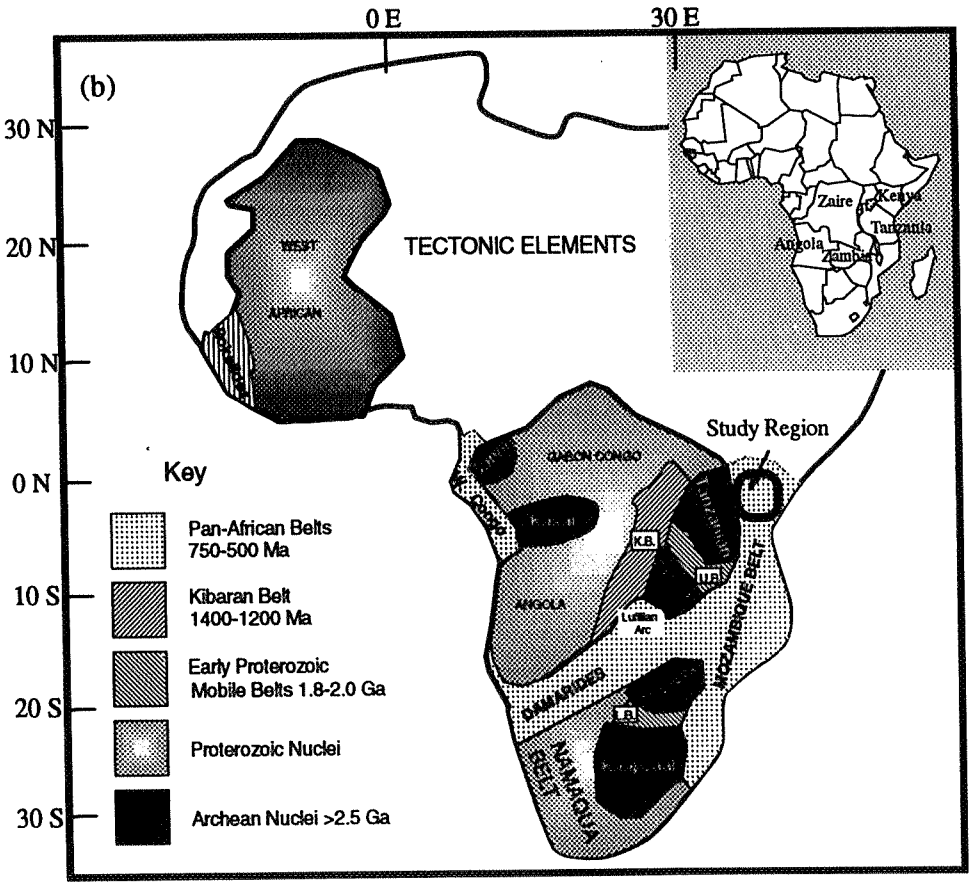
[1] Manuscript received April 18, 1995; accepted Sept 27, 1995.

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[Journal of Geology, 1996, volume 104, p. 131–142] © 1996 by The University of Chicago. All rights reserved. 0022-1376/96/10402-0009$01.00


750 Ma
W. Africa
Baltica
Amazonia & Rio Plata
0 S
Siberia
1100-1300 Ma orogenic belts
Laurentia
30 S
Australia
Pole of Opening at 700 Ma
E. Antarctica
Kal.
Congo
60 S
India
Madagascar

(a)


0 E
30 E
(b)
30 N
20 N
10 N
0 N
10 S
20 S
30 S
TECTONIC ELEMENTS
Zaire
Kenya
Tanzania
Angola
Zambia
Study Region
Key
Pan-African Belts 750-500 Ma
Kibaran Belt 1400-1200 Ma
Early Proterozoic Mobile Belts 1.8-2.0 Ga
Proterozoic Nuclei
Archean Nuclei >2.5 Ga
ANGOLA
K.B.
Lufilian Arc
MOZAMBIQUE BELT
NAMAQUA BELT

within Rodinia suggests that the east Antarctic craton and Australia begin to separate from western Laurentia. Dalziel (1992) envisions a wide proto-Pacific ocean between western Laurentia and Australo-Antarctica as the Mozambique Ocean closed between the Congo craton and the impinging Australo-Antarctic landmass. Rifting events in eastern Laurentia culminated in continental separation between 600 and 550 Ma (Bond et al. 1984; Meert et al. 1994*a*; Williams and Hiscott 1987). Thus, according to Dalziel (1992), a second short-lived supercontinent existed between 650 and 550 Ma with a united Gondwana juxtaposed against the eastern margin of Laurentia. Both Dalziel (1992) and Stern (1994) treat West Gondwana as distinct cratonic elements (e.g., Rio Plata, Amazonia, Congo, and West African cratons) assembled during a series of Late Proterozoic orogenic episodes. East Gondwana is generally treated as a coherent unit from at least 1300 Ma until its breakup in the Mesozoic, but the concept of a contiguous East Gondwana is questioned by Kröner (1993), Meert and Van der Voo (1994), Meert et al. (1995), and Wilson et al. (1995). They argue that the assembly of East Gondwana was accomplished through a polyphase accretion of cratons between 800 and 550 Ma.

Despite debates regarding the timing of Gondwana assembly, the Congo craton was located in a critical central position within Gondwana. Therefore, detailed knowledge of the Congo craton's position through time is crucial to unraveling the complex history of Gondwana assembly. We present new paleomagnetic and geochronologic data from the Mozambique Belt immediately adjacent to the Congo craton and examine the implications of these data for Gondwana assembly.

East Africa Orogeny (EAO) and Younger Events

Key et al. (1989) detailed a sequence of orogenic events in the Mozambique Belt of Kenya. The oldest of these, the Samburuan-Sabachian Orogeny at ≈820 Ma, is ascribed to plate collision between the Tanzanian craton and an eastern "Kibaran" craton. Recent U-Pb zircon ages from East Africa and the Arabian-Nubian shield suggest that the local Samburuan-Sabachian Orogeny was part of the larger 650–800 Ma East Africa Orogeny (Stern 1994). A post-collisional Baragoin-Barsaloian episode between 620–570 Ma resulted in further deformation and isotopic resetting in the rocks of the Mozambique Belt in central Kenya. Several post-Barsaloian dikes intrude the granulite-grade Mozambique Belt rocks in Kenya and were themselves metamorphosed to greenschist-facies between 570–530 Ma (Key et al. 1989). This greenschist-facies metamorphic episode has a counterpart granulite-facies metamorphic event (550 Ma) outboard of the East Africa Orogen in parts of Antarctica, southern India, Madagascar, and Sri Lanka, raising the possibility that some of East Gondwana was amalgamated significantly after the EAO during a younger and distinct collisional event (Stern 1994; Shiriashi et al. 1994; Kröner 1993).

Paleomagnetic Study

The Sinyai metadolerite dike, roughly 10 km long and 150 m wide, intrudes the Mozambique Belt rocks in central Kenya (figure 2). The dike is discordant to, and its intrusion post-dates, the Baragoin-Barsaloian metamorphic fabric in the region. A younger, greenschist-facies metamorphic episode produced sericitization in the host gneisses and extensive biotite growth in the Sinyai dike. Attempts to date the Sinyai and related dikes produced ages ranging from 593 to 528 Ma, consistent with their presumed age based on structural relationships (Key et al. 1989). We collected a total of 50 samples of metadolerite and host gneisses for paleomagnetic study along two profiles in the dike. Samples were collected from the center of the dike to the contact with the metamorphic rocks it intrudes and through the central part of the dike. Samples were drilled in the field using a gasoline-powered drill or alternatively, oriented hand samples were collected and later drilled in the laboratory at the University of Michigan. Individual cores were then cut to standard 22 mm samples for measurement in a 2-G Enterprises cryogenic magnetometer.

The intensity of natural remanent magnetization in these rocks ranged between 10^{-4} and 2 A/m. Thermal and alternating field (AF) demag-

Figure 1. (*a*) The supercontinent Rodinia reconstructed to its 750 Ma disposition according to Dalziel (1992). The presumed pole of opening between western Laurentia and East Gondwana is shown off the tip of the Kalahari craton. The cratons that will ultimately join to form Gondwana encircle Laurentia along its (present-day) eastern, western, and southern margins. (*b*) Regional map of Africa showing the outlines of the cratonic nuclei which comprise the present-day continent and the orogenic belts separating them. K.B. = Kibaran Belt; U.B. = Ubendian Belt; L.B. = Limpopo Belt.

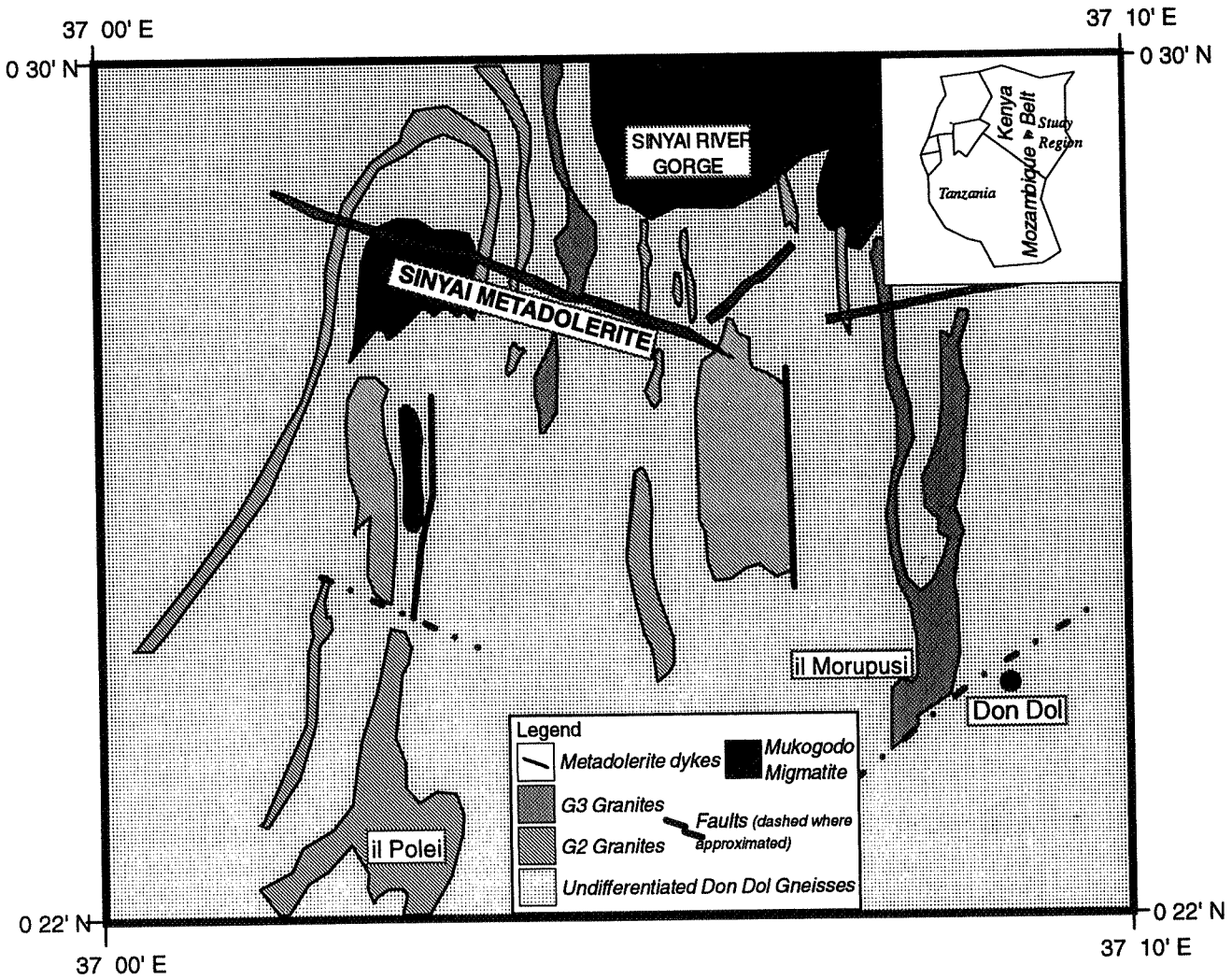


Figure 2. Geologic map of the central Kenya Mozambique orogenic belt showing the location of the Sinyai metadolerite dike.

netization on a pilot selection of samples demonstrated that thermal treatment resulted in a clearer definition of magnetic components within the samples. Demagnetization results were plotted in orthogonal vector diagrams (Zijderveld 1967), and characteristic remanence components were calculated using least-squares methods (Torsvik 1992).

Most samples show two components of magnetization (figure 3). The first, low-temperature component is removed by 500°C. The declinations for this component show considerable scatter between northwest and northeast with consistently shallow inclinations. We attribute this direction to a viscous overprint of recent origin superimposed on the ancient remanence. The second, high-temperature vector is directed to the northeast and shallow-up or to the southwest and shallow-down (figure 4; table 1). The southwest-down direction predominates in most samples and yields a mean direction of D = 244°, I = 20° (N = 28 samples, k = 20 $\alpha 95$ = 6°). The northeast and upwardly directed vectors yield a mean of D = 55° I = 19° (N = 14 samples, k = 20, $\alpha 95$ = 9°). The overall mean direction obtained after reversing the NE-up directions is D = 241°, I = +20° (N = 42 samples, k = 20, $\alpha 95$ = 5°). The dual-polarity magnetizations is the Sinyai metadolerite are approximately reversed with respect to each other and yield a positive reversals test with a classification of **Rc** (McFadden and McElhinny 1990). Rock magnetic tests were carried out to determine the carriers of the magnetic remanence. Low-Ti magnetite is suggested by the unblocking temperature spectra (figure 3); isothermal remanence acquisition studies (IRM, figure 5*a*,*b*) and Curie temperature experiments (figure 5*c*,*d*).

Cores of rock taken near the contact with the Sinyai dike were collected in an attempt to verify

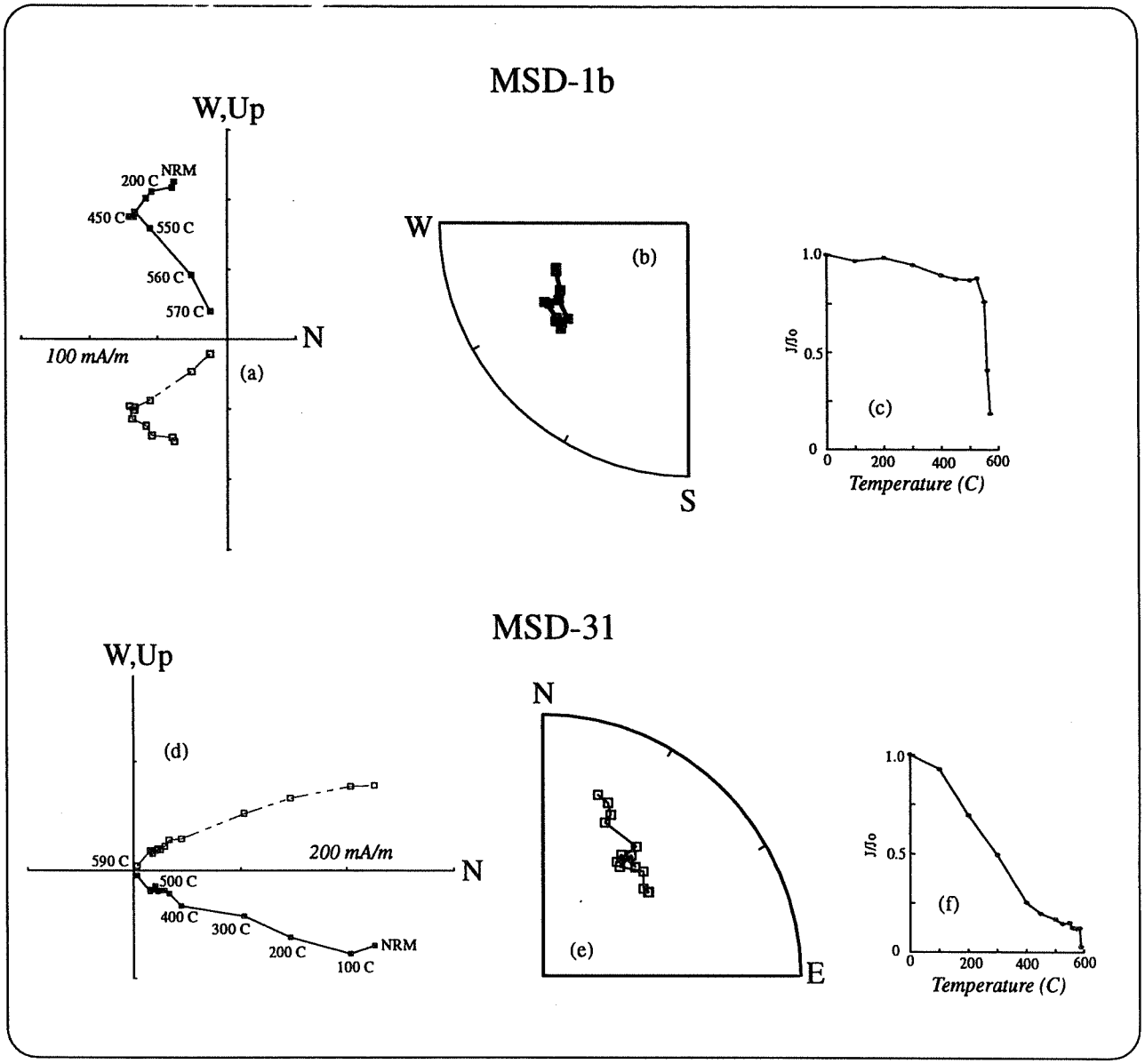


Figure 3. Paleomagnetic results for samples from the Sinyai metadolerite dike shown as (*a,d*) orthogonal vector diagrams, (*b,e*) equal-angle stereoplots, and (*c,f*) unblocking temperature spectra. Sample MSD-1b exhibits the predominate SW-down directed remanence vector and sample MSD-31 exhibits the NE-upwardly directed remanence

the primary nature of the remanence through a baked-contact test. The host rocks showed a single northerly and shallowly inclined direction indistinguishable from the Earth's present dipole field direction for central Keyna. This observation, combined with the often severe weathering of the host rock samples and greenschist-facies metamorphism of the dike rocks, preclude a definitive baked contact test for the Sinyai dike; however, we present the following arguments for the secondary nature of the Sinyai remanence.

The paleomagnetic pole for the Sinyai metadolerite falls at 29°S, 319°E (south pole. $\delta p = 3°$, $\delta m = 5°$). In general, a paleomagnetic pole calculated from directional data in a single dike is regarded as a virtual geomagnetic pole (VGP) because the magnetization recorded by the rocks is unlikely to have averaged secular variation. Key et al. (1989) estimate that the Sinyai dike was emplaced post-Barsaloian (≤570 Ma) and then suffered greenschist-facies metamorphism shortly after emplacement (i.e., between 570–530 Ma). This metamorphic event resulted in extensive biotite overgrowth in the Sinyai dike, alteration of the dike's primary mineralogy, and likely resetting of the original magnetization. Because this remagnetiza-

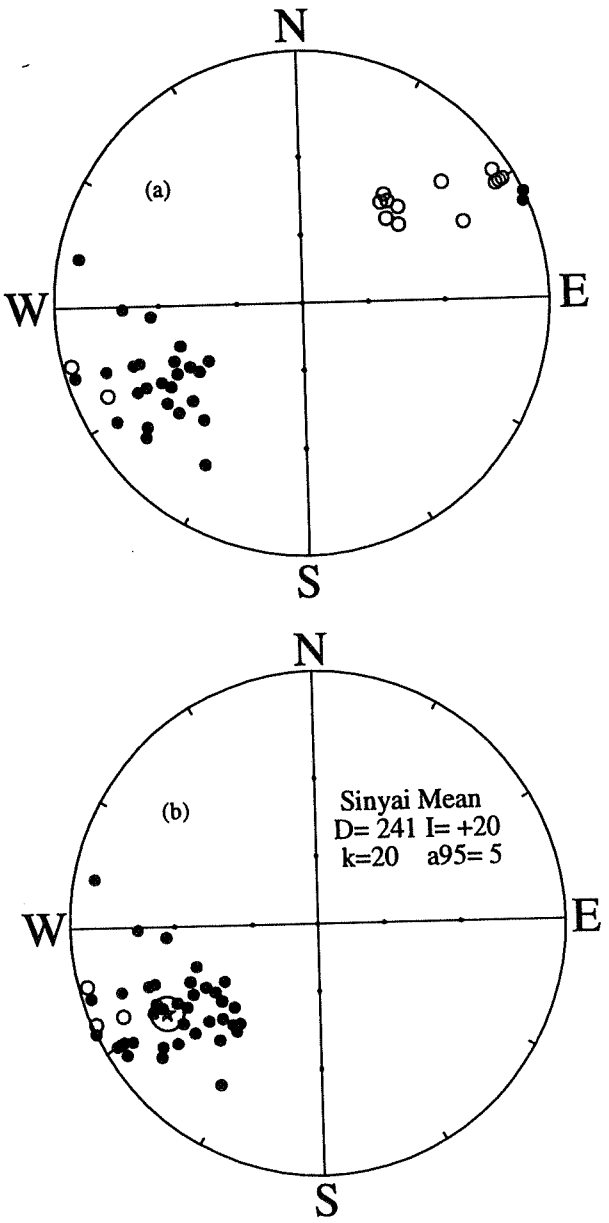


Figure 4. Stereoplot of the sample characteristic directions calculated from the Sinyai dike showing dual-polarities. (*b*) Stereoplot of the sample directions and mean direction obtained by reversing the NE-upwardly directed samples.

Table 1. Paleomagnetic Results

Polarity	Number Samples	Declination (°)	Inclination (°)	kappa (k)	α95 (°)
SW Directions	28	244	+20	20	6
NE Directions	14	055	−19	20	9
Combined	42	241	+20	20	5
Paleomagnetic Pole		29°S	319°E	δp = 3° δm = 5°	

tion occurred over at least one field reversal, we feel that the paleomagnetic pole calculated for the Sinyai dike has averaged secular variation and has a magnetization age dating near the time of biotite growth during metamorphism. We have no way to evaluate completely the possibility of post-emplacement rotation and tilting of the Sinyai dike. Nevertheless, we do note that the metamorphic fabric of the country rock intruded by the dike is concordant with the regional metamorphic fabric, suggesting that no local rotation of the dike has taken place (see Key et al. 1989 for description).

Age of the Sinyai Pole

In an effort to determine the age for the paleomagnetic pole from the Sinyai dike, we separated biotite grains from the dike and dated them using the $^{40}Ar/^{39}Ar$ method. We used biotite grains taken directly from the same cores we used in the paleomagnetic study. Specifically, biotite was removed from sample MSD-31 that yielded a NE-up direction and sample MSD-1 that carried the SW-down direction (figure 3). Following separation, individual grains of biotite were irradiated at the University of Michigan's Ford Phoenix reactor. The samples were then stepwise-degassed using an Ar-laser system, and the argon isotopic ratios (masses 40–36) were measured on a Mass Analyzer Products 215 mass spectrometer operating in a static mode. Neutron flux gradients were monitored using the Mmhb-1 standard (Alexander et al. 1978) using an age of 520.4 Ma. Samples consisted of single grains of biotite, and replicates (splits) were run to determine any grain-to-grain variability. Extraction line blanks were run every five steps, and all peaks are blank-corrected. Further technical details of the $^{40}Ar/^{39}Ar$ methodology are described in Meert et al. (1994*b*). Analytical data for the individual samples/splits are given in table 2.

The computed age spectra for the biotite grains are depicted in figure 6*a* and *b*. We calculated plateau ages when a sample released more than 50% of the gas over three or more successive steps with overlapping ages at the 2σ level (all errors reported herein are 2σ) and a mean squares weighted deviates (MSWD) cutoff value of ≤2.0. Where the age spectra did not yield a plateau because the MSWD values were too high, we added all the gas fractions over the imperfect plateau region to arrive at an integrated age. Replicate samples from MSD-1 yield identical plateau and integrated ages of 550 ± 2 Ma. Total gas ages for the two splits gave similar ages of 548 ± 4 Ma and 550 ± 2 Ma, respectively. Sample MSD-31 yielded a younger plateau

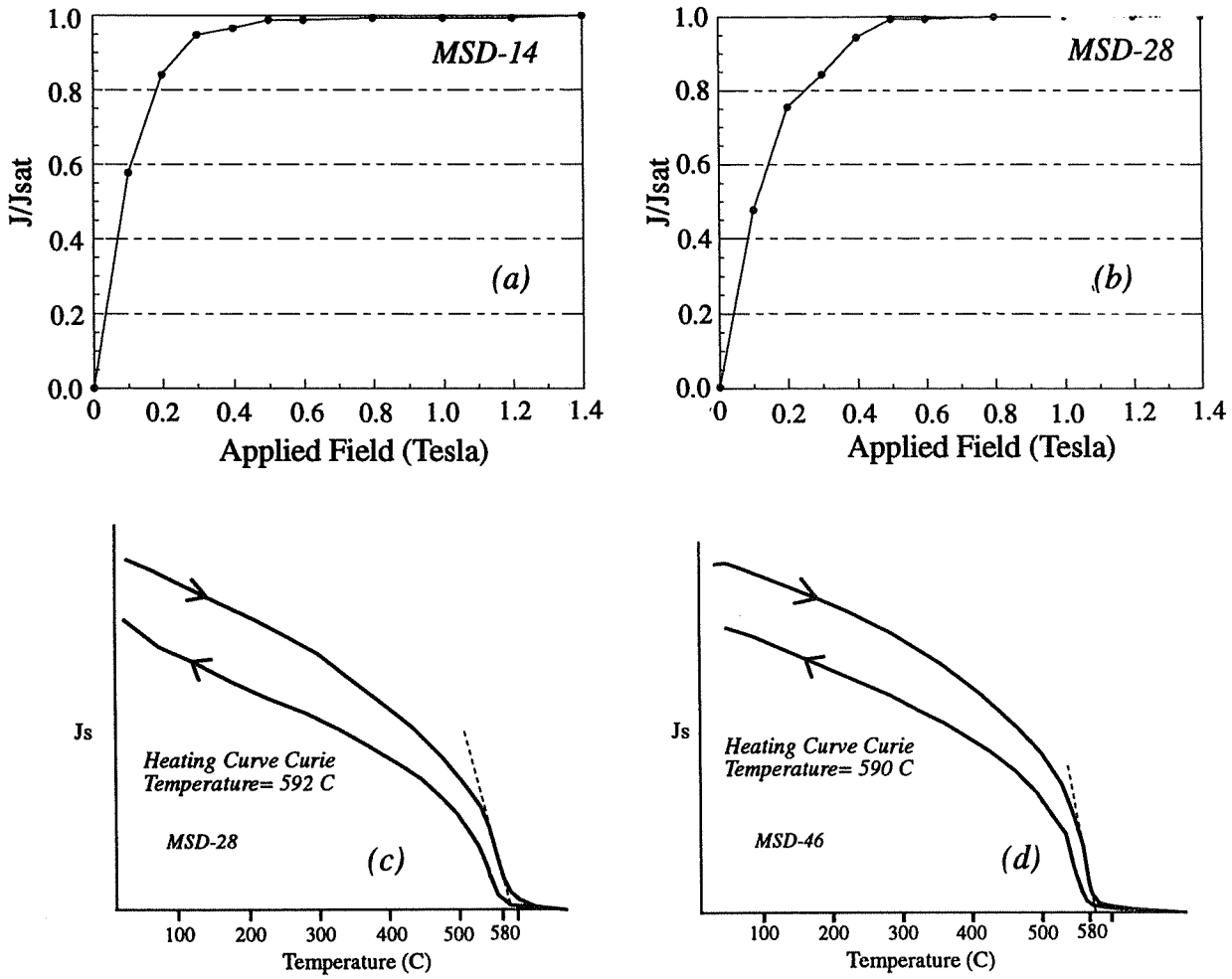


Figure 5. (*a,b*) Isothermal remanence acquisition (IRM) on two samples of the Sinyai metadolerite that show saturation near 0.4 Tesla and (*c,d*) Curie temperature analyses of Sinyai metadolerite samples that exhibit Curie temperatures near 590°C.

age of 543 ± 2 Ma. These age differences between samples that carry opposite polarity magnetizations, while slight, support our contention that the magnetization carried by the Sinyai dike averaged secular variation during protracted remagnetization between 570-530 Ma. Collectively, the argon data suggest an age of 547 ± 4 Ma for the Sinyai pole, and therefore the biotite cooling age corresponds closely to the age of the magnetization.

Discussion

The Sinyai metadolerite pole augments the available database for Gondwana in the interval 550–510 Ma. Table 3 lists the available poles from Gondwana (depicted in figure 7*a*) with ages known to ±6%, α95 (or A95) <20° and Q-value (Van der Voo 1990) >3. We have applied similar criteria in our other models because poles of lesser reliability do

Table 2. $^{40}Ar/^{39}Ar$ Results

Sample Number	J-Value ‡Error	Total Gas Age (Ma) ‡2σ	Plateau (P) or Integrated (I) Age (Ma)	Gas used for Age (%)
MSD-1a (split 1)	.0067616 ± .000008	547 ± 4	550 ± 2 (P)	98
MSD-1a (split 2)	.0067616 ± .000008	550 ± 2	550 ± 2 (I)	98
MSD-31	.0067086 ± .000007	541 ± 2	543 ± 2 (P)	98

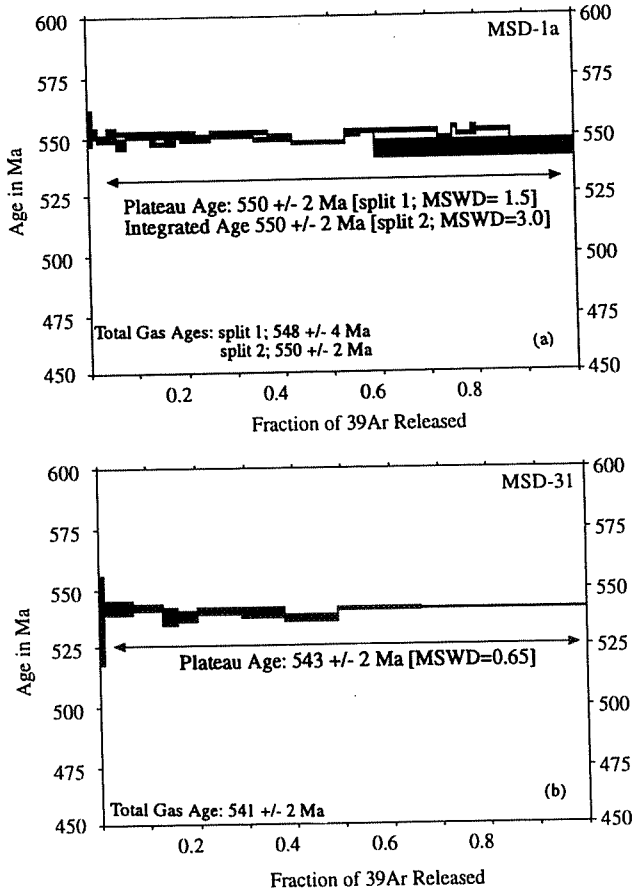


Figure 6. (*a*) Outgassing age spectra for single biotite grains removed from paleomagnetic sample MSD-1a that yield identical plateau and integrated ages of 550 ± 2 Ma and (*b*) outgassing spectra for a single biotite grain removed from sample MSD-31 that yields a plateau age of 543 ± 2 Ma.

not provide useful tests of continental reconstructions (Meert et al. 1995; Van der Voo and Meert 1991). The Sør Rondane pole of East Antarctica (Zijderveld 1968) was previously assigned an age of 490 Ma; however, Grunow (1995) suggested that the geochronologic data from the intrusions favors a 515 Ma age for this pole, and we have included the result in our compilation. There are no poles from the South American cratons that fit our criteria. There are a few paleomagnetic poles from the Kalahari craton for this interval (McWilliams and Kröner 1981; Kröner et al. 1980); however, the age assignments for the multiple magnetization directions reported in those studies preclude an unambiguous interpretation of the results (Van der Voo and Meert 1991). In fact, previous estimates for the timing of Gondwana amalgamation relied heavily on these Kalahari craton results and led to the conclusion that East and West Gondwana were not joined until ≈520 Ma (Li and Powell 1993; Powell et al. 1993). The apparent polar wander path in figure 7*a* shows a relatively coherent swathe of Gondwana poles from the 547 Ma Sinyai pole toward the northwest coast of Africa supporting a united Gondwana by at least 550 Ma. This track also places the glacial deposits in West Africa dated between 556 ± 12 and 520 Ma at high southerly latitudes as West Africa moved toward the south pole during this interval (Bertrand-Sarfati et al. 1995).

Any model for the formation of Gondwana must incorporate an explanation for both the 650–800 Ma East Africa Orogeny (Stern 1994) and the ubiquitous 550 Ma metamorphic events observed throughout East Gondwana. U-Pb zircon ages from East Africa date granulite-facies metamorphism and, presumably, crustal thickening due to continent-continent collision around 700 Ma (Maboko et al. 1985, 1990; Muhongo and Lenoir 1994). These ages are consistent with estimates provided by Key et al. (1989) and Stern (1994) for continent-continent collision in East Africa. Kröner (1993) and Shiriashi et al. (1994) provide U-Pb zircon age evidence for granulite-grade metamorphic events in Sri Lanka, Madagascar, and parts of Enderby Land (Antarctica) at 550 Ma. Deformation in southern India and western Australia is also tentatively dated at 550 Ma, although the exact tectonic significance of these ages is not clear (Harris and Beeson 1994; Cruse et al. 1993). Collectively, these data can be interpreted to support the idea that the assembly of the eastern (Australia and most of East Antarctica) and central (Africa, India, Madagascar, Sri Lanka, parts of Enderby Land and the Annandagstopanne region of Dronning Maud Land) parts of Gondwana occurred during two separate orogenic events (Meert et al. 1995).

Meert et al. (1995) note that the pre-725 Ma paleomagnetic poles from India, Australia, and the Congo craton are scattered, supporting a post-725 Ma consolidation of these cratons. They attribute the 650–800 Ma East Africa Orogen to a collision between the IMSLEK blocks and the Congo craton (figure 7*b*; IMSLEK = southern India, Madagascar, Sri Lanka, parts of Enderby Land and Dronning Maud Land of Antarctica and the Kalahari craton). The later 550 Ma orogeny, resulting from the collision of Congo-IMSLEK with Australia and the bulk of the East Antarctic craton, is given the name Kuunga Orogeny (Kuunga is the Swahili word meaning to unite). It is our contention that the coheren

Table 3. Paleomagnetic Poles from 550–510 Ma

Pole	Q-Value	Age (Ma) ± Error	Pole Latitude	Pole E Longitude	A95 or α95	Rotated to Gondwana
Pan-African Belts adjacent to Congo Craton						
Sinyai Dolerite[a]	5	547 ± 04	29°S	319°	05°	Fixed
Ntonya Ring[b]	4	522 ± 13	28°N	345°	02°	Fixed
East Antarctica						
Sør Rondane Intrusions[c]	3	515	28°S	010°	03°	12°N, 013°E
India						
Bhander and Rewa s.s.[c]	3	550 ± 30	51°S	037°	11°	23°S, 333°E
Purple Sandstone[d]	3	530 ± 15	28°S	032°	12°	05°S, 348°E
Jutana Fm.-Pakistan[e]	4	530 ± 12	21°S	051°	11°	14°S, 005°E
Salt Pseudomorph's[f]	4	523 ± 24	27°S	034°	5°	06°S, 350°E
Central Australia						
Mean Pole[h]	6	550 ± 15	45°S	346°	13°	12°S, 343°E
Mean Pole[h]	6	530 ± 15	36°S	005°	18°	02°N, 351°E
Mean Pole[h]	6	510 ± 15	26°S	022°	18°	17°N, 001°E

Notes. Gondwana Euler poles from de Wit et al. (1988; to rotate to a fixed Africa); India 29.6°N, 36.1°E, −56.8°; Australia 24.6°S, 60.4°W, +49.9°; Antarctica 2.4°S, 327.3°E, +55.4°.
[a] This study.
[b] Briden et al. 1993.
[c] Zijderveld, 1968.
[d] McElhinny et al. 1978.
[e] McElhinny 1970.
[f] Klootwijk et al. 1986.
[g] Wensink 1972.
[h] Compiled from Li and Powell 1993.

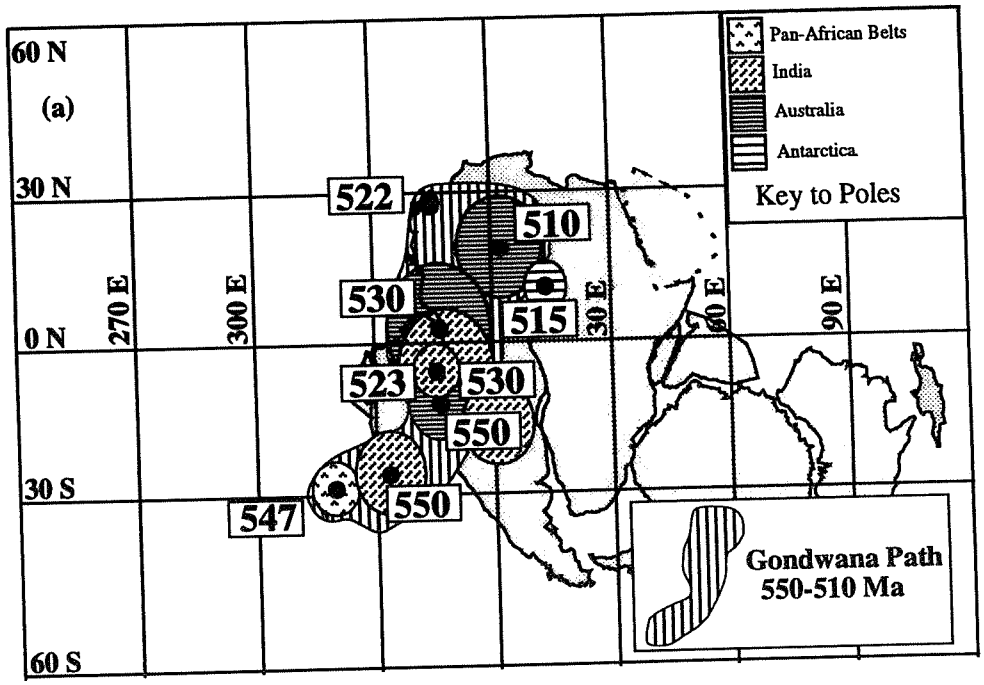
swathe in figure 7*a* represents the time after final amalgamation of the Gondwana cratons. This interpretation requires that East Gondwana, as envisioned in most previous studies, was not a coherent landmass until 550 Ma. Therefore, the previous terms of "East" and "West" Gondwana, referring to specific long-lived continental configurations, should be abandoned in Late Proterozoic models if this interpretation is correct. Our model can ultimately be tested further by continued efforts to obtain new paleomagnetic, geochronologic and structural data from the Gondwana continent.

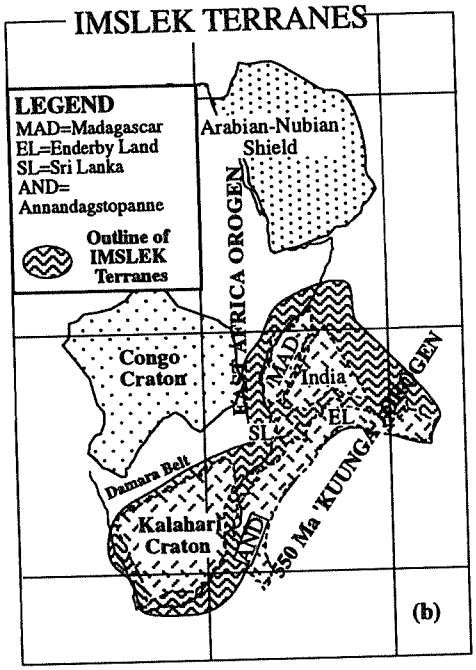
Summary and Conclusions

We report a new paleomagnetic pole from the Sinyai metadolerite dike in central Kenya at 29°S, 319°E ($\delta p = 3°$, $\delta m = 5°$). The remanence directions from the Sinyai samples are dual-polarity and likely resulted from a protracted remagnetization dating from the time of greenschist metamorphism in central Kenya. The effects of this metamorphism included extensive biotite overgrowth in the Sinyai dike, which is dated by $^{40}Ar/^{39}Ar$ methods at 547 ± 4 Ma. This pole, when combined with other reliable paleomagnetic poles from Gondwana, reveals a coherent apparent polar wander path for the cratons of east and central Gondwana beginning at 550 Ma. The 550 Ma age corresponds to the age of granulite-facies metamorphism in southern India, Madagascar, East Antarctica, and Sir Lanka, yet clearly post-dates the 700 Ma granulite-facies metamorphism in East Africa. Collectively, the paleomagnetic and age data suggest the formation of east and central Gondwana by two separate orogenic events; the East Africa Orogeny between 650–800 Ma (Stern 1994) and the Kuunga Orogeny at 550 Ma.

ACKNOWLEDGMENTS

The authors wish to thank Robert Munyua for his able field assistance while collecting the Sinyai samples, Celina Uranga for help measuring the samples; Harald Walderhaug at the University of Bergen (Norway) for providing us with the results of Curie temperature runs, Jayanti Patel at the University of Nairobi for hospitality and arranging to get the samples out of Kenya, Pieter van Dongen of Groundwater Survey Kenya Ltd. for arranging a driver and field vehicle, and Chris Hall and Marcus Johnson for their help with the argon analyses. We





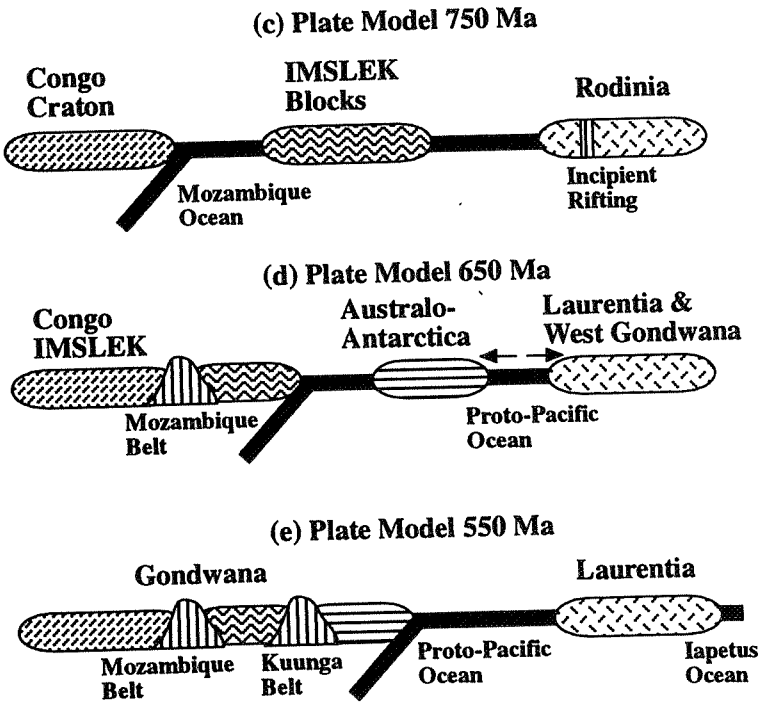


Figure 7. (*a*) Apparent polar wander path for Gondwana using the Sinyai pole and poles listed in table 3 rotated to African coordinates. The path forms a relatively coherent swathe from the Sinyai pole toward the northwest coast of Africa supporting a united east and central Gondwana at 550 Ma. (*b*) Outline of the proposed IMSLEK blocks and the two orogenic belts that may represent the sites of continental collisions during the amalgamation of Gondwana. The plate cartoons (*c,d,e*) show the hypothesized sequence of events that lead to the formation of Gondwana at 550 Ma.

also thank Ian Dalziel and an anonymous reviewer for throughtful editorial comments. The Schürmann Fund of the Netherlands and the Division of Mines & Geology-Kenya supported the fieldwork, whereas the laboratory work was funded by the Division of Earth Sciences, National Science Foundation, Grant EAR 92-05815.

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